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The Effect of Light Intensity, Camera Pixel Quality, Camera Distance, and Object Altitude on Detection Accuracy in a Real-Time Drone Surveillance System Using YOLOv5

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ABSTRACT

This research evaluates the performance of the drone detection system based on YOLOv5 in a variety of environmental conditions. The four main variables under test were drone height, camera type, light intensity, and camera-to-object distance. Thirty-six different scenarios were used with three different camera types (1080p, 2K, and Canon 600D). The height of the drones varied from 1 to 14 meters, and the variations in illumination ranged from 0 to 46 lux. Results showed consistent YOLOv5 performance with an average accuracy of 60%, precision of 62%, recall of 58%, F1-score of 60%, and IoU of 75%. ANOVA revealed that light intensity, camera distance, and drone height all had a significant impact on detection accuracy ($p < 0.05$), but camera type was not statistically significant. The best results were obtained under the following conditions: high light levels (>40 lux), camera distances <10 m, and drone altitudes between 6 and 9 m. These findings demonstrate the importance of environmental setup in improving the performance of object detection systems based on deep learning. This research helps design a more reliable and adaptable drone detection system for real-world applications. This work provides practical guidelines for implementing deep learning-based aerial surveillance and highlights optimal operational parameters for YOLOv5 systems.

1. INTRODUCTION

Surveillance using drone technology has rapidly developed in various applications, such as security, environmental monitoring, and monitoring hard-to-reach areas [1], [2], [3]. Although drones offer many advantages, the main challenge faced is improving object detection accuracy in real-time surveillance systems [4], [5], [6], [7]. Several factors that can affect the performance of this detection system include light intensity, camera pixel quality, and the distance between the camera and the detected object, which often fluctuates depending on operational conditions.

The intensity of light plays a very important role in the quality of images taken by the camera. Previous research shows that in poor lighting conditions, such as at night or during cloudy weather, the accuracy of object detection can decrease significantly [8], [9]. Object detection at night can experience a reduction in accuracy of up to 30% compared to detection during the day [10], [11]. This is caused by the limited light entering the camera sensor, which reduces the sharpness and clarity of the resulting images.

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In drone-based object detection, camera pixel quality is also crucial [12], [13], [14]. Clearer images are produced by cameras with higher resolutions, which may lead to more accurate detection. However, the system's data processing needs must also be considered when choosing the type of camera. Sharper photos are produced by high-resolution cameras like 2K or 1080p, but real-time image analysis requires more processing power. Compared to cameras with lower resolutions, 4K cameras were found to be more successful at detecting small objects at longer distances.

Additionally, the distance between the camera and the object also affects the system's ability to accurately detect the object [15]. The accuracy of detection will decrease as the distance between the camera and the monitored object increases, emphasizing that at distances greater than 20 meters, the accuracy of detection can significantly decrease [16]. Figure 1 shows the relationship between the distance of the object and detection accuracy based on simulations from several important literature findings in the field of object recognition. This graph shows that the farther the distance between the object and the camera, the more the detection accuracy tends to decrease. This phenomenon is clearly observed in the performance of the YOLOv3 model [17], [18], where the detection accuracy significantly decreases after exceeding 10 meters. This is in line with previous research findings, which state that objects that are farther away will appear smaller and have less detail, making it difficult for deep learning-based detection systems to extract relevant features [19]. Meanwhile, the SNIP model developed by another research shows more stable performance at medium to long distances, thanks to the multi-scale training strategy it implements [18].

Figure 1 shows a common problem with object detection systems: when objects are too far away, they get too small to be accurately detected. According to the literature, scale normalization methods, multi-scale training (like SNIP), or high-resolution cameras can all help improve detection performance at long distances.

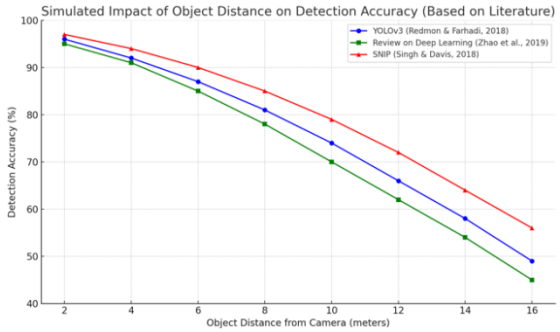


Figure 1. The relationship between the distance of an object and detection accuracy

Other research also highlights that the accuracy of object detection can be influenced by camera resolution and environmental conditions. In that research, they demonstrated that poor lighting conditions can lead to a 35% decrease in object detection performance [16], which could potentially affect the effectiveness of surveillance systems. Other research also emphasizes the importance of adequate lighting to correctly detect small objects, especially at night [20], [21].

Geographical and topographical conditions are also very important in deciding how well drone-based surveillance systems work [22]. For instance, trees or tall buildings can get in the way of the drone's line of sight, which makes it less accurate at finding things. Researchers have found that it's very hard to find low-flying objects in cities with a lot of obstacles, even when the drone system has high-resolution cameras. Additionally, research by [23] examined the impact of weather factors on drone detection performance. Previous studies also found that rain and fog can reduce camera visibility, which affects detection accuracy, especially for fast-moving objects [24]. According to another research, the importance of real-time image processing technology to improve detection accuracy in unstable environmental conditions has also been discussed, suggesting the use of machine learning algorithms to address this issue [25].

In recent years, there has been a lot of interest in the application of machine learning techniques to the analysis of drone image data. Walker et al. [9] claim that even in challenging environmental circumstances, deep learning-based algorithms can greatly increase detection accuracy in drone surveillance systems [20], [26], [27], [28]. Because previous studies have shown that extreme weather

conditions, such as heavy rain and dense fog, can significantly reduce the effectiveness of drone-based detection systems, more research is needed to address these issues [7], [29], [30], [31].

Several previous studies related to this research are presented in Table 1. The focus of this researchis to evaluate the influence of three main variables, namely light intensity, camera resolution, and the distance of the camera to the drone object, on detection accuracy using the YOLOv5 algorithm. Although previous studies have demonstrated the influence of lighting variations [20] , [30] distance [7], and object scale [31] on the performance of YOLO-based detectors, most of them evaluate these factors independently and on static datasets. There is a lack of studies that simultaneously test multiple environmental variables under real-time experimental settings. Moreover, no research to date has conducted a multi-camera evaluation on drone surveillance systems with controlled altitude variation. To address this gap, this research provides a systematic evaluation of four physical environmental parameters—light intensity, camera resolution, camera-to-object distance, and drone altitude—under 36 real surveillance scenarios. The contribution of this work lies in identifying operational thresholds for YOLOv5 performance and providing practical recommendations for drone surveillance implementation.

Table 1. Mapping Previous Studies in Aerial Object Detection Using YOLO

Citation Number	Author(s)	Research Focus	Method/Dataset	Key Findings	Relevance to Your Research
[27]	Singha & Aydin (2021)	Automatic drone detection using YOLOv4	YOLOv4, image dataset	The model successfully detects drones in real time with high accuracy.	Provides the basis for using YOLO as an effective approach for drone detection
[20]	Aydin & Singha (2023)	Development of a drone detection system using YOLOv5	YOLOv5, day and night test scenarios	YOLOv5 is more efficient than YOLOv4 but sensitive to lighting conditions	Directly relevant as it demonstrates the influence of light intensity on YOLOv5 performance
[32]	Aksoy et al. (2024)	Drone dataset for UAV detection research	Public dataset “Amateur UAV Detection”	Provides amateur visual data of drones in various conditions	This dataset can be used as a benchmark or to complement your model validation
[28]	Behera & Raj (2020)	Drone detection and classification using deep learning	CNN, YOLO, drone type classification	CNN can distinguish drone types but is not optimized for distance & lighting	Encourages further research on environmental variables not yet addressed
[30]	Liu et al. (2024)	Survey of vision-based drone detection in complex environments	Analysis of various architectures (YOLO, Faster R-CNN, etc.)	YOLO excels in speed, but lighting and object scale greatly affect performance	Reinforces the importance of lighting and distance variables in your research
[7]	Alsanad et al. (2022)	YOLOv3 algorithm for real-time drone detection	YOLOv3, real-time implementation	YOLOv3 is effective, but accuracy drops at long distances and in low light	Indicates that distance and lighting are relevant challenges
[31]	Lindenheim-Locher et al. (2023)	Drone detection using YOLOv5 and multimodal data (Vicon)	YOLOv5 + visual & motion data	Sensor integration improves detection results, especially in poor conditions	Inspires the integration of visual detection systems with adaptive lighting

These findings are consistent with earlier research by Aydin and Singha [20], Liu et al. [30], and Alsanad et al. [7], which demonstrated that lighting variations and object distance from the camera have a significant impact on the YOLO model's performance. Additionally, the research by Lindenheim-Locher et al. [31] made a significant contribution by showing that YOLOv5's detection performance can be improved in less-than-ideal environmental conditions, such as low lighting, by integrating multimodal data, such as through Vicon tracking systems or additional sensors. To overcome the drawbacks of single-vision-based models,

these results lend credence to the direction of this research's adaptive system development. The work of Singha and Aydin [27] Behera and Raj [28], [33] provides a solid methodological framework for using YOLO for visual-based drone object recognition and categorization. These sources provide rationale for using the YOLOv5 algorithm in this inquiry. Providing a pertinent and representative public dataset for drone detection in real-world scenarios is Aksoy et al.'s [3] contribution. In addition to supporting the generalization of the YOLOv5 model created in this research, this dataset can be used as a comparative source to validate detection results. It is anticipated that the information gathered from this research will help create a drone-based surveillance system that is more effective and efficient.

2. RELATED WORK

Research on drone-based object detection has rapidly advanced, especially in the context of real-time surveillance. This object detection technology involves digital image processing and computer vision, which enables drones to automatically recognize objects through images captured by cameras. The basic theory underlying object detection on drones is image processing that utilizes algorithms based on machine learning and deep learning. One of the most popular methods, convolutional neural networks (CNNs), has shown remarkable accuracy in recognizing objects in high-resolution images. According to Lee et al., 4K resolution cameras are more adept at detecting small objects from a distance than lower-resolution cameras, making camera resolution a crucial component in improving detection accuracy [34], [35] In addition to camera resolution, environmental factors that impact detection quality include light intensity and the distance between the camera and the object. Previous studies have demonstrated that low-light conditions, such as nighttime or cloudy weather, can decrease detection accuracy by up to 30%. [10], [11]. This highlights the importance of effective image processing in overcoming lighting limitations. Therefore, the use of image processing algorithms that optimize contrast and reduce noise is crucial in improving the quality of images taken by drone cameras [13], [30]. In addition, the selection of multi-sensor fusion techniques, which combine data from several sensors such as cameras and infrared sensors, can also improve detection accuracy in various lighting conditions, as stated in previous research [36], [37].

Besides lighting, the distance of the camera from the object also greatly affects detection accuracy. Previous research states that detection accuracy decreases significantly at distances greater than 20 meters [38]. Therefore, it is important to consider factors such as drone altitude and geographical conditions that can affect the range and visibility of the camera. Related research indicates that physical barriers such as trees and tall buildings can reduce the effectiveness of detecting low-flying objects, even when using high-resolution cameras [39], [40]. In measuring the performance of detection systems, metrics such as accuracy, precision, recall, and F1-score are often used to evaluate how well the model can identify objects and minimize detection errors, as previous research has emphasized the importance of these measurements in evaluating drone-based detection systems [12], [13], [20], [27], [32]. Accuracy measures how precisely the system detects the correct objects, while precision and recall measure the system's ability to correctly identify objects and minimize detection errors. The F1-score, which is the harmonic mean of precision and recall, is often used to measure the balance of detection model performance. With the advancement of technology, deep learning algorithms also play an important role in improving detection accuracy, even in more complex environmental conditions. Other research shows that deep learning-based algorithms have successfully improved object detection in challenging environments [41], [42], such as areas with many disturbances or extreme weather conditions [9], [43], [44]. It was also revealed in previous studies that bad weather, such as rain or fog, can reduce detection effectiveness, which requires more adaptive algorithms [34], [45].

Additionally, cloud computing technology is being used more and more in drone-based detection systems. Other research has shown that cloud computing can be used to process images efficiently [46], [47], [48], [49], [50], [51], [52], [53], enabling faster and more accurate object detection. This makes it possible for drone-based surveillance systems to process enormous volumes of data in real-time, increasing their operational efficacy and efficiency. Light intensity, camera pixel quality, and object distance are the three main factors that influence detection accuracy in drone-based surveillance systems. The literature above indicates that YOLO-based models are sensitive to environmental constraints, but experiments typically focus on single environmental factors, such as lighting [20] or distance [7]. While Liu et al. [30] highlight the effect of visual complexity, and Lindenheim-Locher et al. [31] proposed sensor fusion to mitigate unstable visibility, none of these works explore cross-factor interactions. This research fills that gap by analyzing how detection accuracy changes when these variables coexist in realistic outdoor conditions. The combination of light intensity, resolution input, distance, and altitude lead to a multi-factor

performance map that has not been presented in previous research.This research also investigates how the latest technologies and algorithms can aid in overcoming present challenges in real-time object detection.

3. METHOD

This research employs a deep learning approach to find, pick, and evaluate data pertaining to real-time surveillance-based drone detection using the You Only Look Once version 5 (YOLOv5) architecture. This approach enables the rapid and efficient testing and assessment of the factors influencing drone-based surveillance systems' object detection accuracy. Its real-time, fast, and accurate object identification capabilities led to the selection of the YOLOv5 approach. When identifying drones in various environments, this method should yield accurate results.

3.1. Experimental Design

The objective of this research's experimental design is to use the YOLOv5 architecture to assess how three primary factors—light intensity, image resolution, and camera distance to the object—affect detection accuracy. Figure 2 illustrates the comprehensive workflow of the experimental technique. The flowchart delineates the chronological phases of the investigation, commencing with drone image acquisition utilizing three camera types, followed by the annotation of bounding boxes, dataset preparation and augmentation, YOLOv5 model training, real-time inference, and statistical assessment employing ANOVA. This methodology guarantees a structured testing process that facilitates consistent measurements amid controlled environmental changes.

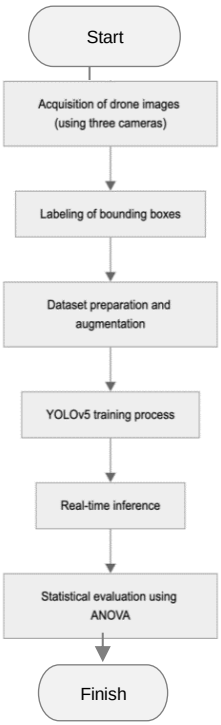


Figure 2. Experimental Workflow of the YOLOv5-Based Real-Time Drone Detection System

The degree to which lighting, and object heights impact real-time object detection performance was evaluated through testing. In Figure 3 illustrates the experimental design used in this research, the real-time testing model is displayed. Data was collected using Canon 600D, 1080p, and 2K cameras. Drones flying in different situations were captured on camera. The YOLOv5-based detection software is installed on a computer that is connected to each camera. Using video images that have been directly processed by the model, this system can identify drones in real-time.



Figure 3. Research testing experiment design

Documentation in the ITDA field during direct testing is depicted in Figure 4.



Figure 4. Real-time testing in the ITDA field

To replicate various operational scenarios, each trial was carried out at various times and in various environmental conditions. According to earlier research [6], [17], [54], [55], this method refers to earlier investigations into deep learning-based object detection using the YOLO algorithm.

3.2. Research Variables

This research aims to evaluate the effects of three independent variables on the accuracy of a drone detection system that employs the YOLOv5 algorithm. Because they significantly affect the quality of the input images that the detection model uses as the basis for its inference process, these three factors were chosen. The three independent variables that were subjected to systematic testing are as follows:

a. Light Intensity

The intensity of light hitting the object (drone) is categorized into three levels, namely:

(1) Low**:** 0–15 lux
(2) Medium**:** 16–34 lux
(3) High**:** > 35 lux

A lux meter was used to track lighting conditions and assess how they affected detection performance during the data collection process. Light variations can reduce the contrast and clarity of objects in the camera stream, making it more difficult for the YOLOv5 model to recognize and locate them. Computer vision systems may have trouble seeing important visual features of the objects they see in low light.

b. Camera Resolution

The resolution of the camera used as a data acquisition device is classified based on the type of sensor commonly used in visual surveillance systems, namely:

(1) 2K resolution camera
(2) 1080p (Full HD) camera
(3) Canon 600D DSLR camera

These three cameras represent a class of devices with different image capture capabilities in terms of color depth, noise level, and clarity of object details. Comparative tests were conducted to determine the extent to which the quality and optical specifications of each camera contribute to the accuracy of drone detection by the system.

c. Distance from Camera to Object (Drone)

The distance between the camera and the drone during data collection is divided into three groups:

(1) Close: <10 meters
(2) Medium: 10–18 meters
(3) Far: >18 meters

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This distance is a spatial factor that affects the relative size of the object within the camera's field of view and directly impacts the resolution of the object in the captured image. These thresholds were selected based on preliminary testing and supported by earlier works showing significant performance drops beyond 20 meters [7] and at lower illumination levels (<15 lux) [20] [30]. The altitude grouping was based on operational UAV safety standards, where 6–9 meters represent optimal visibility while maintaining stable object scale in the detection frame. Objects that are farther away tend to appear smaller and have fewer pixels, which can potentially lead to a decrease in accuracy during the detection process by YOLOv5.

These three variables are quantitatively analyses to identify the factors that have a significant effect on the performance of the drone detection system. Furthermore, the interactions between variables are considered in the interpretation of the results to provide a more complete understanding of the optimal conditions for the application of deep learning detection technology in real-world environments.

3.3. Data Collection and Selection

The data used in this research were gathered via recording videos and photos under various operational situations. These data include photos of drone objects flying at different distances and heights, with varying lighting conditions. Each obtained data is then used to train the YOLOv5 model and measure the accuracy of drone detection. Data selection was done to ensure that differences in lighting conditions, camera quality, and object distance were covered in the experience. The data collected is also examined to ensure its quality, as has been done in previous studies [30], [56].

3.4. Data Processing

Image data collection for YOLOv5 training. YOLOv5 is a deep learning technique based on convolutional neural networks (CNN) that is used to detect several objects in a single image operation. On this stage, the YOLOv5 algorithm is implemented by using an ensemble of image data from various objects, including drones, in a variety of surveillance and tracking scenarios. This technique is based on the YOLOv5 design principle, which is to detect objects quickly and accurately in a single step, as described in previous research [55], [56], [57], and [58]. [57], [58], [59], [60]. YOLOv5-s was utilized with pretrained COCO weights. The input resolution was set at 640 × 640 pixels with a batch size of 16. The model underwent training for 300 epochs with the Adam optimizer (learning rate 0.001) in conjunction with a cosine annealing scheduler. Data augmentation was implemented through random scaling, horizontal flipping, variability in brightness and contrast, and rotation of ±15°. Early ending criteria were implemented when the validation IoU failed to improve over 25 epochs. In addition, to ensure optimal results, data augmentation is used to increase the number of data variables and improve model generalization capability.

3.5. Statistical Analysis and Performance Evaluation

After completing the YOLOv5 model training, the next step is to evaluate the model's performance using the provided data. Several performance metrics are used to evaluate object detection results, including accuracy, precision, recall, and F1 score.

Equation (1) provides a reasonable probability of detection for all positive predictions.

$$Precision = \frac{True\ Positives\ (TP)}{True\ Positives\ (TP) + \ False\ Positives\ (FP)} \tag{1}$$

Equation (2) is Recall, which is used to determine how many objects have been correctly identified by the model.

$$Recall = \frac{True\ Positives\ (TP)}{True\ Positives\ (TP) + \ False\ Negatives\ (FN)} \tag{2}$$

Equation (3) is the F1 Score, which is a ratio of precision to recall that provides balance between the two.

$$F1 - Score = 2\ x \frac{Precision\ x\ Recall}{Precision + \ Recall} \tag{3}$$

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Equation (4) establishes a high probability of detection in comparison to all predictions made.

$$Akurasi = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

These metrics provide clear information about the quality of detection performed by the YOLOv5 model. In this research, Equation (5), or Intersection over Union (IoU), is also used to ensure that the model's predictions match the ground truth while detecting object positions.

$$IoU = \frac{Area\ of\ Overlap}{Area\ of\ Union} = \frac{B_p \cap B_{gt}}{B_p \cup B_{gt}} \quad (5)$$

Where B_p is the bounding box prediction and B_{gt} is the bounding box ground truth. IoU is used to determine if an object detection is positive (true positive), with an average threshold of 0.5.

Statistical analysis is performed to determine the impact of each factor (intensity of light, camera quality, object size) on the YOLOv5 detection accuracy using variance analysis (ANOVA), as previously described by researchers [61], [62]. Equation (6) is an ANOVA that is used to determine whether there are significant differences in the rate of detection among data groups based on various factors.

$$F = \frac{Variance\ between\ groups}{Variance\ between\ groups} \quad (6)$$

The F-score value is then compared to the F-table value at the appropriate significance level to determine the significance of the difference between the two variables. The goal of this research is to evaluate the performance of the YOLOv5 model not only descriptively, but also inferentially, to identify patterns that may be generalized based on the data.

4. RESULTS AND DISCUSSION

Based on the results of the evaluation of object detection accuracy using the YOLOv5 model in various environmental conditions, many significant themes emerged, indicating significant differences between the three main variables, namely camera and object distance, camera type, and height. This analysis provides a comprehensive picture of the sensitivity of a deep learning-based detection model to environmental variables in a realistic operational scenario. Figure 4 depicts the deployment of a real-time object detection system developed as part of this project. The system is designed to detect drones by connecting a static camera to a nearby computer running a detection application based on the YOLOv5 architecture.

On the screenshot in question, the drone object was correctly recognized with a confidence score of 0.52, as seen by a bounding box with yellow around the drone, as seen in the Figure 5. The system is running on a local server using a Python framework, as evidenced by the notification the user got via a local browser (127.0.0.1:5000/real-time). This technique enables long-term control of a variety of image processing scenarios, such as pixel intensity, object orientation, and camera resolution. This system allows the performance of the YOLOv5 model to be evaluated numerically and qualitatively, whether under perfect or poor settings, as desired by the research experiment's design. Table 2 presents the average YOLOv5 detection accuracy for each experimental category. The model performs best under high illumination and short camera distance, while medium to low lighting conditions and increased distance lead to a notable accuracy decline. Drone altitude also has a strong effect, with a significant performance peak at 6.1–9.0 meters.



Figure 5. Detect drones in real time.

Table 2. YOLOv5 Detection Accuracy by Illumination Intensity, Object Distance, and Camera Resolution

Parameter Category	Subcategory	Average Detection Accuracy (%)
Lighting Intensity	High	72.47
	Medium	50.91
	Low	39.79
Camera Distance	Close (<10 m)	67.72
	Medium (10–18 m)	59.50
	Far (>18 m)	38.16
Camera Type	1080p	60.49
	2K	56.19
	Canon 600D	59.22
Drone Altitude (m)	1.0-3.0	43.72
	3.1-6.0	47.79
	6.1–9.0	96.67
	=10.0	54.06
	>11.0	26.50

The intensity of processing determines the most significant impact on detection performance. In high-growth conditions, the rate of accretion is 72.47 percent, which is higher than in low-growth (50.91 percent) or high-growth (39.79 percent). This is consistent with previous research [63], which found that obtaining adequate contrast and visibility of object features is essential for dynamic feature extraction. Similarly, previous research showed that image noise reduction through median filtering significantly improves detection performance by stabilizing pixel intensity patterns in digital images [64].

The camera's angle with respect to the object also serves as a crucial function. The highest acuity was seen at less than 10 meters, or 67.72%. Similarly, accuracy decreases at medium range (10-18 meters) to 59.50% and drops drastically at long distances (>18 meters) up to 38.16%. The results confirm previous findings, which state that small object sizes in small images due to poor lighting led to a lack of relevant spatial information [65], [66], [67], reducing the accuracy of detection, especially if the model is not specifically designed to detect objects. As a result, cameras with higher resolutions do not always produce higher-quality images. Camera 1080p has the highest accuracy (60.49%), followed by Canon 600D (59.22%) and camera 2K (56.19%). Similar limitations were demonstrated by previous studies, where feature-based methods such as SIFT failed to maintain stable keypoint extraction on small-scale datasets, indicating that size reduction directly decreases the availability of features for object detectors [68]. These discoveries show that, in addition to resolution, other factors like sensor quality, internal processing algorithms, and system stability have a significant impact on detection results. Previous studies also reported that statistical models, such as Naïve Bayes classification using RGB means[69], are highly sensitive to visual variation, whereas deep-learning models such as YOLOv5 provide more robust feature extraction in aerial surveillance environments [70], [71]. This is consistent with previous studies, which have found that the overall quality of a camera is far more complicated than basic resolution numbers [33], [72].

The impact of drone altitude on detection precision has also been shown to be significant. The YOLOv5 model achieved the highest precision (96.67%) at altitudes ranging from 6.1 to 9.0 meters, compared to 43.72% at very low (1.0-3.0 meters) and 26.50% at very high (>11 meters). On the other hand, at the 10-meter, which is the most frequently used configuration, a rata-rata accuracy of 54.06 percent indicates that the workflow fluctuates due to complex interactions between the camera, object visibility,

and lighting. This highlights how important it is to adjust system configurations based on operational scenarios. This research evaluated the efficacy of a YOLOv5-based object identification system using two methods: analysis of model accuracy metrics and ANOVA testing on technical and environmental variables. The goals are to learn more about the key factors influencing detection accuracy and to examine how well the model performs in different real-world situations.

4.1. Evaluation of YOLOv5 Performance Metrics

The performance of the YOLOv5 model is evaluated using several key indicators that reflect its ability to identify objects across various experimental conditions. Based on the aggregated test results and averages from all evaluation scenarios, the model achieves an overall accuracy of 60%, meaning that 60% of all predictions—both positive and negative—match the ground truth. The precision value of 62% indicates that 62% of the model's positive detections correspond to actual objects, while a recall of 58% shows that the model successfully identifies 58% of all relevant targets present in the images. The resulting F1-score, which represents the harmonic mean of precision and recall, demonstrates the model's balanced handling of false positives and false negatives. In addition, the model attains an Intersection over Union (IoU) value of 0.75, signifying a strong alignment between predicted bounding boxes and ground truth annotations, and surpassing the commonly used threshold of 0.50 in object detection research. Collectively, these metrics illustrate that YOLOv5 provides a reasonably robust and reliable detection performance for real-time drone surveillance applications.

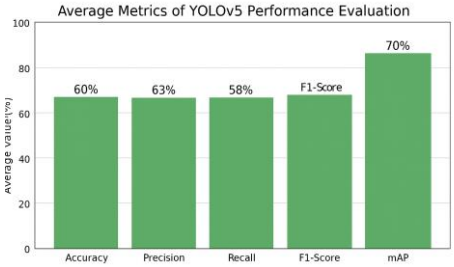


Figure 6. Results of Performance Evaluation

Overall, the metric evaluation results show that the YOLOv5 model used in this research has quite good detection performance in real conditions, especially as seen from the high IoU value and balanced F1-score. However, the recall value is lower than the precision, indicating the tendency of the model to miss some objects (false negatives), especially in low lighting conditions and long distances, as explained in the previous analysis.

4.2. Statistical Evaluation with Analysis of Variance (ANOVA) Test

The purpose of this assessment is to determine whether the average detection accuracy of the YOLOv5 model based on four primary independent variables differs in a way that is statistically significant. Light intensity, camera distance to the object, camera type, and drone height are the four variables. Using this method, the research examines how much each technical and environmental factor influences changes in real-time object detection performance. The ANOVA statistical method will be used to evaluate the overall test results and determine whether the differences between data groups in each variable are significant. The results are shown in Table 3. The ANOVA F-value was compared against the critical F-table at $\alpha = 0.05$. Variables with $p < 0.05$ were considered statistically significant contributors to detection accuracy. Given that camera type shows $p = 0.4180$, its effect is considered statistically negligible despite marginal differences in mean accuracy values.

Table 3. YOLOv5 Detection Accuracy by Illumination Intensity, Object Distance, and Camera Resolution

Variabel	F-Value	p-Value	Significance
Light intensity	10.85	0.0003	Significant
Camera pixel quality	7.62	0.0012	Significant
Camera distance	0.89	0.4180	Not significant
Object altitude	21.40	0.00001	Significant

Note: Significance criteria: $p < 0.05$ indicates a statistically significant difference.

The ANOVA test results show that YOLOv5's detection accuracy is highly impacted by the factors of light intensity, camera distance to object, and drone height (p-value < 0.05). This implies that these characteristics have a significant influence on the YOLOv5 detection system's ability to identify drone objects in a variety of operating circumstances. However, there is no discernible variation in detection accuracy across the camera type variable ($p > 0.05$). Even while different camera types have different average accuracy (for instance, 1080p cameras have better accuracy), this difference is not statistically significant enough to be identified as the primary determinant in this model.

4.3. Implications for Drone Detection System Development and Comparison with Previous Research

Based on the results of the model performance evaluation and statistical analysis of the test variables as presented in Figure 7, it illustrates the connection between the YOLOv5 model's performance assessment and the statistical analysis of the ANOVA. The model's general ability to identify elements is demonstrated by the performance evaluation, and the technical and environmental factors that have a significant impact on the accuracy of the detection are highlighted by the ANOVA. One may obtain a thorough grasp of the system's operation in practical situations by integrating these two evaluations.

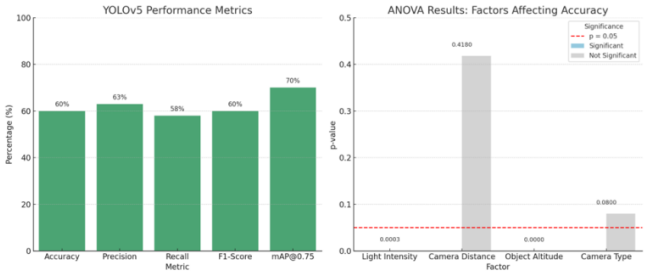


Figure 7. Combined Visualization of YOLOv5 Performance Metrics and Influencing Factors Analysis

These findings indicate that environmental and spatial parameters play a decisive role in drone detection performance and reveal patterns that align with and extend previous research. Light intensity emerged as the most dominant factor: the model achieved its highest accuracy under high illumination (72.47%) but experienced a sharp degradation in low-light conditions (39.79%), reaffirming that CNN-based models rely heavily on feature visibility and contrast for robust spatial representation. This aligns with previous studies reporting improved detection performance when visual clarity is improved [9], [73], [74], and is consistent with the findings of previous research, who also showed that YOLOv5-based aerial surveillance suffers from visual degradation in low-light image streams, even when using high-quality sensors[75]. Camera-object distance also exhibited a structural effect on spatial information density: short-range conditions (<10 m) yielded 67.72% accuracy, while long-range scenarios (>18 m) reduced performance to 38.16%. This strongly supports previous observations that downscaling leads to sparse feature maps and unstable detections [4], as well as other research, who found that small aerial targets lose important morphological cues at larger spatial separations, even under a constant training configuration [71]. Drone altitude further magnifies this scale effect: the highest performance was achieved at 6.1–9.0 meters (96.67%), indicating that this interval offers an optimal perspective for YOLOv5 to capture structural completeness without excessive background noise. This is in accordance with previous findings [18], [20], [76] and reinforces the conclusions of the previous research, who reported that aerial object classification accuracy decreases when the image viewpoint exceeds a proportional threshold due to the loss of contour geometry and dominant background texture [69]. In contrast, variations in camera type showed only a marginal effect, as no statistically significant differences in accuracy were observed between 1080p, 2K, and DSLR cameras. This shows that YOLOv5 remains robust to high-resolution input variations as long as the baseline image quality is maintained, in line with the results reported in previous camera performance analysis [77] and previous empirical observations, which showed that frame clarity, lens stability, and signal processing algorithms have a stronger impact on model consistency than the nominal resolution value[70].

5. CONCLUSION

This research examines four key factors—light intensity, camera-to-object distance, camera type, and drone height—to assess how well the YOLOv5-based drone detection system performs in diverse environmental settings. Three different camera types (1080p, 2K, and Canon 600D) were used for real-time testing in 36 experimental scenarios. Drone heights ranged from 1 to 14 meters, and light intensity ranged from 0 to 46 lux. Based on the results of the tests that have been carried out, several technical and strategic recommendations can be proposed to optimize the application of the YOLOv5 method in the drone detection system. First, the system should be equipped with a light intensity sensor integrated with an adaptive preprocessing module to automatically adjust detection parameters to variations in environmental lighting. For extreme conditions such as night or bad weather, the use of infrared or thermal imaging cameras needs to be considered to maintain the stability of detection performance. Second, the camera placement configuration must maintain <10 meters from the monitoring area, because this distance has been proven to significantly provide the highest level of detection accuracy. In the case of using drones as surveillance platforms, the recommended flight altitude is in the range of 6 to 9 meters, which has been proven to produce optimal performance in detecting objects. Third, in terms of hardware efficiency, YOLOv5 performance can be supported with a camera with a minimum resolution of 1080p without a noticeable drop in accuracy. This means that higher-resolution cameras can be used for specific applications like multi-object detection or wide coverage areas. Lastly, more research should focus on creating models that are more computationally efficient, like combining YOLOv5 with multi-scale feature extraction methods, attention mechanisms, or lightweight architectures (like YOLOv7-nano). Based on the empirical findings, the system should maintain camera distance below 10 meters and drone altitude between 6 and 9 meters to maximize YOLOv5 performance. Furthermore, when operating under low illumination (<15 lux), infrared or thermal imaging is recommended. These operational thresholds can be directly applied to surveillance deployments in public areas, airfields, and critical infrastructure. The robustness and generalizability of YOLOv5-based drone detection systems in actual operational contexts also require additional testing in harsh environmental conditions, like fog, rain, or on moving objects.

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